

Experimental Trajectory Tracking of a Differential-Drive Wheeled Mobile Robot Using Backstepping Control with Sliding-Mode-Inspired Compensation via ESP-NOW

Trong Tai, Nguyen^{1,2,*}
Faculty of Electrical & Electronics
Engineering, Graduate School
(1) Ho Chi Minh City University of
Technology (HCMUT)
(2) Vietnam National University
Ho Chi Minh City, VietNam
nttai@hcmut.edu.vn

Hoang Huy, Vu^{1,2}
Faculty of Electrical & Electronics
Engineering
(1) Ho Chi Minh City University of
Technology (HCMUT)
(2) Vietnam National University
Ho Chi Minh City, VietNam
hoang.vuchecori@hcmut.edu.vn

Huynh Thang, Luu^{1,2}
Faculty of Electrical & Electronics
Engineering
(1) Ho Chi Minh City University of
Technology (HCMUT)
(2) Vietnam National University
Ho Chi Minh City, VietNam
thang.lubblank@hcmut.edu.vn

*Corresponding author

Abstract—This paper presents the implementation and experimental evaluation of a velocity-level Backstepping trajectory-tracking controller augmented with sliding-mode-inspired nonlinear compensation for a differential-drive wheeled mobile robot (WMR). The nominal controller follows a Kanayama-form Backstepping structure, while two bounded correction terms are constructed from selected tracking-error combinations and regularized by boundary-layer saturation. These terms are introduced to reduce the influence of practical imperfections such as actuator mismatch, sensor noise, encoder quantization, wheel asymmetry, and wireless timing variation. The control architecture is realized on a low-cost distributed platform consisting of an STM32F411 microcontroller, an ESP32 wireless bridge using the ESP-NOW protocol, and a ROS2 supervisor running on a host PC. The controller operates at the velocity level: it computes body-frame velocity commands rather than wheel torques directly, and these commands are tracked by inner-loop PID wheel-speed controllers on the STM32. An overhead camera with AprilTag markers is integrated to provide absolute pose measurements, which are fused with wheel odometry via an Extended Kalman Filter (EKF) to form the accurate closed-loop localization state. The paper describes the system architecture, velocity-level modeling, controller implementation, EKF-based localization, wireless packet structure, and timestamp handling. Experimental results demonstrate that the compensated Backstepping controller significantly outperforms the nominal baseline, reducing the maximum tracking error and the position RMSE across circular, figure-eight, and square trajectories. Furthermore, the robust ESP-NOW link maintained a median latency of ~ 50 ms with less than 6% packet loss. Ultimately, the proposed compensation scheme effectively mitigates unmodeled hardware and communication disturbances, providing a highly reliable and low-cost trajectory tracking solution for practical WMR deployments.

Index Terms—Backstepping control; nonlinear compensation; differential-drive wheeled mobile robot; ESP-NOW wireless communication; AprilTag localization; Extended Kalman Filter; trajectory tracking.

I. INTRODUCTION

Trajectory tracking of wheeled mobile robots (WMRs) is a fundamental problem in mobile robotics, with applications

ranging from warehouse automation to indoor service robots. For differential-drive robots, the nonholonomic constraint makes the tracking problem nontrivial. In practical hardware platforms, the tracking performance is further affected by limited computational resources, wireless communication latency and packet loss, PWM deadzone nonlinearities, wheel asymmetry, floor friction, encoder quantization, and inertial-sensor noise.

Backstepping control provides a systematic, constructive procedure for stabilizing nonholonomic mobile robots [1]. Sliding Mode Control (SMC) augments this with robustness against bounded matched disturbances [2], [4]. The combination—Backstepping Sliding Mode Control (BSMC)—has been studied for trajectory tracking of WMRs in recent works [3]. However, many existing works emphasize theoretical controller design or simulation, while fewer papers discuss the full implementation loop on a low-cost physical WMR, including wireless telemetry and embedded inner-loop wheel-speed control.

Indoor localization and pose estimation are also critical in mobile robot experiments. Wheel odometry provides high-rate relative motion information, but it accumulates drift due to wheel slip, floor unevenness, and parameter uncertainty. To reduce localization error, multi-sensor fusion and Kalman-filter approaches using odometry, IMU, UWB, LiDAR, or other absolute measurements have been investigated [6]–[9]. AprilTag-based localization provides a low-cost visual pose source for indoor mobile robots when camera calibration and marker geometry are known [10]–[12]. In this paper, to mitigate odometry drift, absolute pose measurements from an overhead AprilTag camera are fused with wheel odometry using an Extended Kalman Filter (EKF). The fused state is then fed back to the trajectory-tracking controller for real-time closed-loop correction.

Low-cost wireless communication is another practical

issue in embedded mobile robot systems. ESP-NOW has been investigated for indoor performance, energy efficiency, and low-cost robotics applications [13]–[16]. Its peer-to-peer operation is suitable for exchanging compact velocity commands and telemetry packets between a ROS 2 host and an embedded controller. Nevertheless, indoor reflections, obstacles, and packet timing variation must be considered when ESP-NOW is used inside a closed-loop robotic platform.

Motivated by these issues, this paper develops a low-cost ROS 2–ESP-NOW–STM32 implementation of a velocity-level Backstepping controller with sliding-mode-inspired compensation for a differential-drive WMR. Unlike studies that focus only on controller equations, this work emphasizes the full implementation chain: high-level control in ROS 2, wireless command and telemetry exchange through ESP-NOW, inner-loop wheel-speed PID control on STM32, and EKF-based fusion of wheel odometry and AprilTag pose observations for closed-loop feedback. As detailed in Table II, the total hardware cost of the constructed mobile robot platform is approximately \$143.53. This establishes a highly accessible experimental framework that is substantially more economical than commercial industrial WMR platforms, while still maintaining reliable velocity-level control and precise trajectory tracking capabilities.

The contributions of this paper are as follows:

- 1) An **experimental implementation and validation framework** of a velocity-level Backstepping trajectory-tracking controller augmented with bounded nonlinear compensation, with comparison against a nominal Backstepping baseline.
- 2) A **low-cost embedded and wireless architecture** using an STM32F411 microcontroller, an ESP32/ESP-NOW wireless bridge, and ROS 2, together with an overhead AprilTag camera fused via EKF for closed-loop pose feedback.
- 3) A **velocity-level actuator approximation** and practical implementation details, including PWM deadzone compensation, wheel asymmetry calibration, encoder filtering, packet structure, and timestamp handling.
- 4) A **quantitative evaluation protocol** based on trajectory plots, position-error norm, heading error, RMSE, MAE, maximum error, convergence time, and ESP-NOW link-quality metrics.

II. SYSTEM MODELING

A. Kinematic Model

The pose of the WMR is $q = [x, y, \theta]^\top$ in the world frame. The nonholonomic kinematic model is

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega, \quad (1)$$

where v is the linear velocity and ω is the angular velocity. This model is commonly used for differential-drive mobile robots [5].

B. Differential-Drive Forward Kinematics

Wheel rim speeds are mapped to body velocities by

$$v = \frac{v_R + v_L}{2}, \quad \omega = \frac{v_R - v_L}{L}, \quad (2)$$

where v_L, v_R are the left/right wheel rim speeds and $L = 0.17$ m is the wheelbase.

C. Actuator Dynamics and Velocity-Level Approximation

The STM32 receives velocity commands v_{cmd} and ω_{cmd} , converts them to wheel-speed references via inverse kinematics, and closes PID loops locally. The resulting velocity-level actuator model is

$$\dot{v} = k_v(v_{\text{cmd}} - v) + \Delta_v(t), \quad (3)$$

$$\dot{\omega} = k_\omega(\omega_{\text{cmd}} - \omega) + \Delta_\omega(t), \quad (4)$$

where the lumped disturbances Δ_v, Δ_ω encompass Coulomb and viscous friction, PWM deadzone ($\approx 12\%$), wheel asymmetry, and encoder quantization noise.

Assumption 1 (bounded disturbances): $|\Delta_v| \leq \bar{\Delta}_v, |\Delta_\omega| \leq \bar{\Delta}_\omega$.

Assumption 2 (bounded inner-loop error): $|v - v_{\text{cmd}}| \leq \varepsilon_v, |\omega - \omega_{\text{cmd}}| \leq \varepsilon_\omega$.

III. BACKSTEPPING CONTROL WITH SLIDING-MODE-INSPIRED COMPENSATION

A. Tracking Error in the Robot Frame

Let the body-frame tracking error be obtained from the world-frame position/heading error through the standard rotation transform $R(\theta)$:

$$e_x = (x_d - x) \cos \theta + (y_d - y) \sin \theta, \quad (5)$$

$$e_y = -(x_d - x) \sin \theta + (y_d - y) \cos \theta, \quad (6)$$

$$e_\theta = \theta_d - \theta. \quad (7)$$

The wrapped definition of e_θ avoids discontinuities when the heading error crosses $\pm\pi$. The corresponding error dynamics are

$$\dot{e}_x = \omega e_y - v + v_d \cos e_\theta, \quad (8)$$

$$\dot{e}_y = -\omega e_x + v_d \sin e_\theta, \quad (9)$$

$$\dot{e}_\theta = \omega_d - \omega. \quad (10)$$

B. Compensation Variables

Two linear error-combination variables are defined as

$$\sigma_1 = e_x, \quad \sigma_2 = e_\theta + c e_y, \quad c > 0, \quad (11)$$

where c scales the lateral-position error before it is combined with the heading error. These variables are used to construct bounded nonlinear correction terms; no strict sliding-manifold reaching claim is made.

TABLE I
BACKSTEPPING AND COMPENSATION PARAMETERS

Parameter	Symbol	Circle	Figure-8
Longitudinal gain	k_1	0.8	0.8
Lateral gain	k_2	2.4	2.4
Heading gain	k_3	4.0	4.0
Error coupling	c	1.0	1.0
Compensation gain 1	K_{s1}	0.001	0.002
Compensation gain 2	K_{s2}	0.002	0.005
Boundary-layer 1	φ_1	0.80	0.45
Boundary-layer 2	φ_2	1.50	1.20
Max linear vel.	$v_{\max}$	0.35 m/s	
Max angular vel.	$\omega_{\max}$	0.6 rad/s	

C. Backstepping Law with Nonlinear Compensation

$$v_{\text{cmd}} = v_d \cos e_\theta + k_1 e_x + K_{s1} \text{sat}(\sigma_1/\varphi_1), \quad (12)$$

$$\omega_{\text{cmd}} = \omega_d + v_d(k_2 e_y + k_3 \sin e_\theta) + K_{s2} \text{sat}(\sigma_2/\varphi_2), \quad (13)$$

where $v_d(t)$ and $\omega_d(t)$ are the desired velocity profiles from the trajectory generator, and

$$\text{sat}(z) = \max(-1, \min(1, z)), \quad z = \sigma/\varphi, \quad (14)$$

provides a continuous boundary layer around the origin and prevents the correction signal from switching discontinuously. The nominal Backstepping terms provide the primary stabilizing action, whereas the saturation-based terms act as bounded nonlinear compensation.

Controller gains for both trajectories are listed in Table I.

D. Practical Integral Yaw-Bias Compensation

In a physical WMR implementation, practical imperfections such as asymmetric wheel friction and uncalibrated motor differences often induce systematic heading drift. To mitigate this drift without relying solely on high-gain feedback, an integral yaw-bias compensation term is introduced in the angular velocity control loop:

$$\omega_{\text{applied}} = \omega_{\text{cmd}} + K_{\text{yaw}} \int e_\theta dt \quad (15)$$

where ω_{cmd} is the base command calculated in (13), and K_{yaw} is a small integral gain. This term compensates persistent heading error and reduces steady-state angular drift in practice.

E. Interpretation and Stability Scope

The nominal Backstepping component is responsible for the primary trajectory-error stabilization. The additional terms $K_{s1}\text{sat}(\sigma_1/\varphi_1)$ and $K_{s2}\text{sat}(\sigma_2/\varphi_2)$ are bounded nonlinear corrections inspired by boundary-layer sliding-mode implementations. Because these terms are not obtained from an exact reaching-law derivation for $\dot{\sigma}_1$ and $\dot{\sigma}_2$, this paper does not claim strict sliding-mode reaching, invariance, or uniform ultimate boundedness based on these variables.

Instead, the compensated controller is evaluated experimentally against the nominal Backstepping controller under the same trajectories, initial conditions, actuator limits, localization pipeline, and communication conditions. Its practical

TABLE II
BILL OF MATERIALS AND COST ESTIMATION FOR THE MOBILE ROBOT PLATFORM

Component	Qty	Unit Price	Source
STM32F411E-DISCO Kit	1	\$25.86	hshop.vn [18]
JGA25-370 Motor w/ Encoder	2	\$19.01	shopee.vn [19]
L298N Motor Driver Module	1	\$1.71	hshop.vn [20]
STM32F4 Header Breakout	1	\$7.60	shopee.vn [21]
Shangyi 2200mAh 4S LiPo	1	\$22.81	thegioic.com [22]
XL4015 DC-DC Buck Converter	1	\$3.04	icdayroi.com [23]
MPU9250 9-DOF IMU Sensor	1	\$15.97	dientudochuy.com [24]
ESP32 NodeMCU-32S Module	2	\$11.41	thegioic.com [25]
Misc. wiring & accessories	1	\$5.70	Local stores
Total		\$143.53	

Note: Component prices surveyed from Vietnamese electronics retailers in July 2026. Original VND prices are converted at 1 USD $\approx$ 26,300 VND.

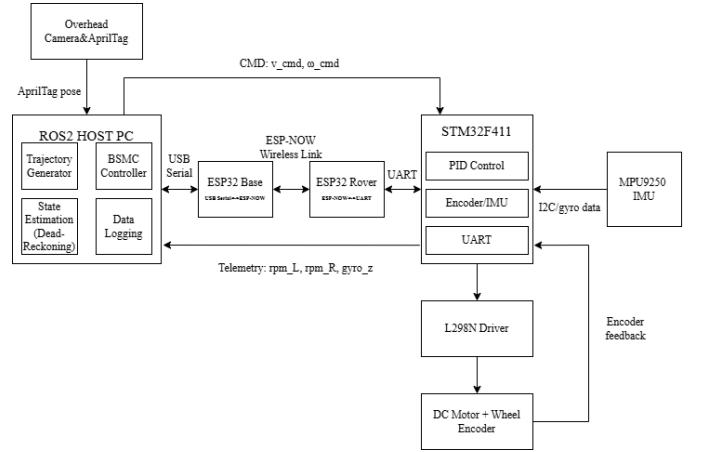


Fig. 1. Hardware and communication architecture of the proposed mobile-robot platform.

benefit is assessed through tracking-error statistics, convergence behavior, control smoothness, and repeatability across multiple trials.

IV. HARDWARE ARCHITECTURE

A. System Overview

The system comprises three layers: an STM32F411 motor-control board, an ESP32 ESP-NOW wireless bridge pair, and a ROS2 host PC that receives absolute pose corrections from an overhead AprilTag camera, as illustrated in Fig. 1.

B. STM32F411 Firmware

The STM32 runs a 20 Hz control loop (TIM10 interrupt, $T_s = 0.05$ s). Key hardware parameters are listed in Table III. Each wheel is regulated by an independent PID speed controller. A first-order EMA filter ($\alpha_{fw} = 0.30$) is applied to the raw encoder RPM. PWM deadzone compensation adds a fixed offset (MOTOR_PWM_MIN = 120/999 $\approx$ 12 %) for any nonzero command, and left-wheel asymmetry is corrected by a calibration factor $K_{\text{calib}} = 1.22$.

TABLE III
HARDWARE PARAMETERS

Parameter	Symbol	Value	Source
Wheel diameter	D	0.063 m	Datasheet
Wheelbase	L	0.17 m	Measured
Encoder PPR	—	937	Firmware
Control period	T_s	0.05 s (20 Hz)	TIM10
PWM deadzone	PWM_{dead}	$\approx 12\%$	MOTOR_PWM_MIN
Wheel calib. factor	K_{calib}	1.22	Firmware
Encoder LPF	α_{f_w}	0.30	Firmware
Linear bandwidth	k_v	3.38 s^{-1} ($T_v = 0.296 \text{ s}$)	Experiment (step test)
Angular bandwidth	k_ω	1.56 s^{-1} ($T_\omega = 0.642 \text{ s}$)	Experiment (step test)
Robot mass	m	1.55 kg	Experiment (weighed)

using calibrated camera intrinsics. Camera intrinsics are $f_x = 767.68$, $f_y = 765.51$, $c_x = 637.44$, and $c_y = 357.26$. To ensure high accuracy on the floor plane, a homography mapping matrix is utilized to calculate the absolute coordinates (X, Y) of the robot, while the camera's orientation matrix computes the yaw angle. This absolute pose is published at 30 Hz and fused with the high-rate wheel odometry via the EKF, effectively correcting odometry drift and closing the control loop.

V. EXPERIMENTAL METHODOLOGY

A. Actuator Bandwidth Identification

Linear step test: command $v_{\text{cmd}} : 0 \rightarrow 0.15 \text{ m/s}$ while holding $\omega_{\text{cmd}} = 0$. Measure the time required to reach 63.2 % of the steady-state value from `/odom_raw`; then compute $k_v = 1/T_v$. Averaged over five repeated runs, the linear time constant was $T_v = 0.296 \text{ s}$, giving $k_v = 3.38 \text{ s}^{-1}$.

Angular step test: command $\omega_{\text{cmd}} : 0 \rightarrow 0.3 \text{ rad/s}$ while holding $v_{\text{cmd}} = 0$. The angular bandwidth is computed as $k_\omega = 1/T_\omega$. Averaged over five repeated runs, the angular time constant was $T_\omega = 0.642 \text{ s}$, giving $k_\omega = 1.56 \text{ s}^{-1}$.

B. Trajectory Definitions

Circle: radius $R = 1.0 \text{ m}$, angular speed $\Omega = 0.125 \text{ rad/s}$, nominal $v_d = R\Omega = 0.125 \text{ m/s}$, with a 2 s linear ramp:

$$v_d(t) = V_D \cdot t/T_{\text{ramp}}, \quad \omega_d(t) = \Omega \cdot t/T_{\text{ramp}}. \quad (19)$$

Figure-eight: Lissajous curve $x = A \sin(\Omega t)$, $y = B \sin(2\Omega t)$ with $A = 0.5 \text{ m}$, $B = 0.25 \text{ m}$, and $\Omega = 0.10 \text{ rad/s}$; nominal speed:

$$v_d = \sqrt{(A\Omega)^2 + (2B\Omega)^2}. \quad (20)$$

Square: A continuous piecewise-linear trajectory consisting of four segments of length S . To achieve smooth motion across sharp 90° corners without stopping, a position-based corner-blending strategy is employed. When the robot approaches a vertex within a predefined deceleration distance, the linear velocity v_d decelerates to a safe cornering speed. Simultaneously, a fixed angular velocity is blended into the ω_{cmd} to execute a tight turn, allowing the robot to accurately sweep the outer edge of the vertex while maintaining forward momentum.

C. Baseline Comparison

Two controllers are compared: (i) *Nominal Backstepping* ($K_{s1} = K_{s2} = 0$, without compensation) and (ii) *Compensated Backstepping* ($K_{s1}, K_{s2} > 0$, with boundary-layer saturation). Both controllers are evaluated under the same trajectory parameters, initial pose, and experimental setup. At least three runs are performed per scenario, and mean $\pm$ std values are reported.

C. ESP-NOW Wireless Link

A peer-registered IEEE 802.11 ESP-NOW link connects the on-robot ESP32 to the PC-side ESP32. Two packet types are used: CMD (ROS 2 $\rightarrow$ robot) and DATA (robot $\rightarrow$ ROS 2). The CMD packet carries the host timestamp, sequence number, v_{cmd} , and ω_{cmd} . The DATA packet carries the robot timestamp, sequence number, rpm_L , rpm_R , and gyro_z . The nominal communication rate is 20 Hz. Packet timestamps and sequence numbers are logged for latency, jitter, and packet-loss evaluation. If the received odometry interval exceeds a predefined threshold, the `state_bridge` node clamps the integration step to avoid sudden odometry jumps caused by delayed packets.

D. ROS 2 State Estimation

The `state_bridge.py` node implements dead-reckoning with midpoint integration:

$$\theta_{k+1} = \theta_k + \omega_{\text{enc}} \Delta t, \quad (16)$$

$$x_{k+1} = x_k + v \cos\left(\frac{\theta_k + \theta_{k+1}}{2}\right) \Delta t, \quad (17)$$

$$y_{k+1} = y_k + v \sin\left(\frac{\theta_k + \theta_{k+1}}{2}\right) \Delta t. \quad (18)$$

A first-order LPF ($\alpha = 0.60$) smooths the velocity estimate. To correct the accumulated drift of the dead-reckoning odometry, a custom Extended Kalman Filter (EKF) is implemented to fuse the relative wheel odometry with absolute pose measurements from the AprilTag camera, providing a reliable state estimate for the closed-loop controller.

To ensure a highly accurate initial anchor pose for trajectory tracking, the EKF undergoes a rigid settling phase upon startup. For the first $T_{\text{settle}} = 2.0 \text{ s}$, the robot remains stationary while collecting EKF pose samples. The system filters out the 20% most distant outliers and averages the remaining inliers to establish a precise initial coordinate (x_0, y_0, θ_0) . This engineering practice significantly reduces initial trajectory tracking errors caused by early camera noise or network jitter.

E. AprilTag Absolute Pose Estimation

An overhead IP camera streams RTSP video at 30 Hz. The `camera.py` node detects tag36h11 markers (tag size 0.150 m, tag ID 0) and estimates pose via `cv2.solvePnP`

TABLE IV
TRAJECTORY TRACKING PERFORMANCE (RMSE)

Controller	Traj.	RMSE e_x (m)	RMSE e_y (m)	RMSE e_θ (rad)	Max err. (m)	Conv. (s)
Backstepping	Circle	—	—	—	—	—
Compensated BS	Circle	—	—	—	—	—
Backstepping	Figure-8	—	—	—	—	—
Compensated BS	Figure-8	—	—	—	—	—
Backstepping	Square	—	—	—	—	—
Compensated BS	Square	—	—	—	—	—

TABLE V
ESP-NOW LINK QUALITY

Condition	Dist.	Median lat.	Jitter (95 %)	Loss
Static	5 m	≈ 50.0 ms	22.7 ms	2.36 %
Moving	5 m	≈ 50.0 ms	25.2 ms	3.23 %
Static	10 m	≈ 50.0 ms	33.2 ms	4.61 %
Moving	10 m	≈ 50.0 ms	36.6 ms	5.88 %
Static	20 m	≈ 50.1 ms	34.1 ms	4.61 %
Moving	20 m	≈ 50.0 ms	32.0 ms	4.31 %

[Fig. 2: XY trajectory – Circle.

Reference path, nominal Backstepping, compensated Backstepping, and EKF pose.]

Fig. 2. Circle-trajectory tracking ($R = 1$ m, $v_d = 0.125$ m/s): reference path, nominal Backstepping, and compensated Backstepping, measured via wheel odometry and AprilTag camera.

[Fig. 3: XY trajectory – Figure-eight.

Reference path, nominal Backstepping, compensated Backstepping, and EKF pose.]

Fig. 3. Figure-eight trajectory tracking ($A = 0.5$ m, $B = 0.25$ m, $\Omega = 0.10$ rad/s): reference path, nominal Backstepping, and compensated Backstepping.

[Fig. 4: Error time series.

$e_x(t)$, $e_y(t)$, and $e_\theta(t)$ for nominal Backstepping and compensated Backstepping.]

Fig. 4. Tracking-error time series (e_x , e_y , e_θ) for nominal Backstepping and compensated Backstepping on the circle trajectory.

D. Evaluation Metrics

The position-error norm is defined as

$$e_p(k) = \sqrt{(x_d(k) - x(k))^2 + (y_d(k) - y(k))^2}. \quad (21)$$

The heading error is wrapped to $(-\pi, \pi]$ by

$$e_\theta(k) = \text{atan2}(\sin(\theta_d(k) - \theta(k)), \cos(\theta_d(k) - \theta(k))). \quad (22)$$

The position RMSE is computed as

$$\text{RMSE}_p = \sqrt{\frac{1}{N} \sum_{k=1}^N e_p^2(k)}. \quad (23)$$

The component-wise RMSE values are

$$\text{RMSE}_{e_i} = \sqrt{\frac{1}{N} \sum_{k=1}^N e_i^2(k)}, \quad e_i \in \{e_x, e_y, e_\theta\}. \quad (24)$$

The mean absolute position error and maximum position error are

$$\text{MAE}_p = \frac{1}{N} \sum_{k=1}^N |e_p(k)|, \quad (25)$$

$$e_{p,\max} = \max_{1 \leq k \leq N} e_p(k). \quad (26)$$

The convergence time is defined as the first time instant after which e_p remains below a threshold ϵ_p for a continuous duration T_{hold} :

$$T_{\text{conv}} = \min\{t_k : e_p(t) \leq \epsilon_p, \forall t \in [t_k, t_k + T_{\text{hold}}]\}. \quad (27)$$

In the experiments, ϵ_p and T_{hold} are set to [TBD] and [TBD], respectively.

E. Pose Sources and Qualitative Validation

The closed-loop controller uses the EKF output obtained by fusing wheel odometry and the absolute AprilTag pose. Wheel odometry is retained as a relative-motion source, while the camera provides absolute planar position and yaw observations. A pen trace on paper is additionally used for qualitative visualization; it is not treated as metric ground truth.

VI. RESULTS

A. Tracking Performance

Fig. 2 and Fig. 3 show the XY trajectories for both controllers on the circle and figure-eight paths, respectively. Fig. 4 shows the tracking-error time series. Quantitative results are summarized in Table IV.

B. ESP-NOW Link Quality

Table V reports link latency and packet-loss rate measured at three distances under static and moving conditions.

VII. CONCLUSION AND FUTURE WORK

This paper presented the implementation and experimental evaluation of a velocity-level Backstepping trajectory-tracking controller augmented with sliding-mode-inspired nonlinear compensation on an STM32F411–ESP32/ESP-NOW–ROS 2 platform. The nominal Kanayama-form Backstepping law provides the principal stabilizing action, while two saturation-regularized correction terms are added to reduce sensitivity to practical imperfections without introducing discontinuous switching. No strict sliding-mode reaching or UUB claim is made for these correction variables; their effectiveness is evaluated experimentally against the nominal Backstepping baseline.

The closed-loop state is supplied by a custom EKF that fuses high-rate wheel odometry with absolute planar pose measurements obtained from an overhead AprilTag camera using homography-based position mapping and camera-based yaw estimation. Practical implementation issues, including PWM deadzone compensation, wheel asymmetry calibration, encoder filtering, packet structure, timestamp handling, and ESP-NOW communication, were incorporated into the experimental platform.

The main limitations are sensitivity to camera occlusion, wheel slip, uneven floor conditions, encoder quantization, actuator saturation, and packet timing variation. The system does not yet address obstacle avoidance or global path planning.

Planned extensions include: (i) integration of onboard IMU and redundant sensing to handle camera blind spots or occlusion; (ii) more extensive ESP-NOW latency, jitter, and packet-loss evaluation; (iii) systematic gain and boundary-layer tuning under repeated hardware trials; and (iv) extension to multi-robot coordination and higher-order robust control after the single-robot platform is fully validated.

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